Problem

A company claims that the **average delivery time is 30 minutes**.

A random sample of **64 deliveries** has an average delivery time of **32 minutes**. The population standard deviation is known to be **8 minutes**.

At a **5% significance level**, test whether the actual average delivery time is different from 30 minutes.

Step 1: Should we use Z-test or T-test?

We know:

- Sample size, $n = 64$
- Sample mean, $\bar{x} = 32$
- Population mean, $\mu = 30$
- Population standard deviation, $\sigma = 8$
- Significance level, $\alpha = 0.05$

Because the **population standard deviation is known**, we use a **Z-test**.

Rule of thumb:
Population σ known → Z-test
Population σ unknown → T-test

Step 2: Define the hypotheses

The company claims:

$$\mu = 30$$

We want to check whether the actual mean is **different** from 30.

Therefore:

$$H_0: \mu = 30$$
$$H_1: \mu \neq 30$$

This is a **two-tailed test**.

Step 3: Calculate the Z-statistic

The Z-test formula is:

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

Substitute the values:

$$Z = \frac{32 - 30}{8/\sqrt{64}}$$

Since:

$$\sqrt{64} = 8$$

we get:

$$Z = \frac{2}{8/8}$$
$$Z = \frac{2}{1}$$

Step 4: Find the critical value

For a two-tailed test at:

$$\alpha = 0.05$$

the critical values are:

$$Z = \pm 1.96$$

We have:

$$\alpha = 0.05$$

And our alternative hypothesis is:

$$H_1 : \mu \neq 30$$

Because of the ≠, this is a **two-tailed test**.

So we divide the significance level between the two tails:

$$\frac{\alpha}{2} = \frac{0.05}{2} = 0.025$$

Therefore:

α/2 = 0.025 α/2 = 0.025

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-1.96 +1.96

There is 2.5% in each tail.

Step 1: Find the area to the LEFT of +1.96

The total probability is:

$$1$$

The right tail is:

$$0.025$$

Therefore the area to the left is:

$$1 - 0.025$$
$$= 0.975$$

So we need to find the Z-value whose cumulative probability is:

$$\boxed{0.975}$$

From the **standard normal Z-table**:

Therefore:

$$\boxed{Z_{critical} = 1.96}$$

Because the test is two-tailed, the other critical value is:

$$\boxed{-1.96}$$

So:

We calculated from the data:

$$Z = 2$$

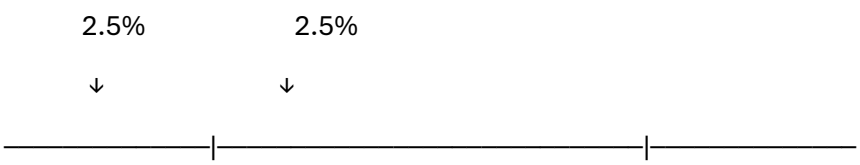
Critical values:

$$-1.96, \ +1.96$$

So our result is:

$$2 > 1.96$$

Therefore, our Z-statistic falls in the **rejection region**.



-1.96 +1.96

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Z = 2

REJECTION REGION

Therefore, Reject H0

Decision rule:

$$| Z |> 1.96 \Rightarrow \text{Reject } H_0$$

Our value is:

$$| 2 |> 1.96$$

Therefore: Reject H0

Step 5: Calculate the p-value

We already calculated:

$$\boldsymbol{Z = 2}$$

For a standard normal distribution:

$$\boldsymbol{P(Z \leq 2) \approx 0.9772}$$

This comes from the Z-table or Python.

So:

$$\boldsymbol{P(Z > 2) = 1 - 0.9772}$$
$$\boldsymbol{= 0.0228}$$

This is the probability in the right tail.

But this is a two-tailed test, so we need both tails.

Because the normal distribution is symmetric:

$$\boldsymbol{p = 2(0.0228)}$$
$$\boldsymbol{p = 0.0456}$$

Approximately:

For $Z = 2$:

$$p \approx 0.0455$$

Compare:

$$p = 0.0455 < 0.05$$

Therefore:

$$\boxed{\text{Reject } H_0}$$

Conclusion

There is statistically significant evidence at the 5% level that the **average delivery time is different from 30 minutes**. Notice that we are not saying the company's claim is definitely false. We are saying that the sample provides sufficient statistical evidence against the null hypothesis.

2. T-Test Numerical Problem

Now let's change one important thing.

Problem

A company claims that the **average customer waiting time is 20 minutes**.

A random sample of **16 customers** gives the following waiting times:

18, 22, 21, 19, 25, 17, 20, 23,

24, 18, 21, 19, 22, 20, 26, 21

At a **5% significance level**, test whether the actual average waiting time is different from 20 minutes.

Step 1: Should we use Z-test or T-test?

Here:

- $n = 16$
- Population standard deviation is **unknown**
- We only have the sample observations.

Therefore, we use a **T-test**.

The important distinction is:

Situation	Test
Population σ known	Z-test
Population σ unknown	T-test
Small sample + σ unknown	T-test is especially appropriate
Large sample + σ unknown	T-test is generally used; with large n, it approaches the Z distribution

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Step 2: Calculate the sample mean

The sample is:

$$18,22,21,19,25,17,20,23,24,18,21,19,22,20,26,21$$

The sample mean is:

$$\bar{x} = \frac{\sum x_i}{n}$$

The sum is:

$$\sum x_i = 336$$

Therefore:

$$\bar{x} = \frac{336}{16}$$
$$\boxed{\bar{x} = 21}$$

Step 3: Define hypotheses

The claimed population mean is:

$$\mu = 20$$

We want to know whether it is **different** from 20.

Therefore:

$$H_0 : \mu = 20$$
$$H_1 : \mu \neq 20$$

Again, this is a **two-tailed test**.

Step 4: Calculate sample standard deviation

Because the population standard deviation is unknown, we calculate the **sample standard deviation** s.

The formula is:

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

For this sample:

$$s \approx 2.449$$

Step 5: Calculate the T-statistic

The T-test formula is:

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

Substitute:

$$t = \frac{21 - 20}{2.449/\sqrt{16}}$$

Since:

$$\sqrt{16} = 4$$

we get:

$$t = \frac{1}{2.449/4}$$
$$t \approx 1.633$$

Therefore:

$$\boxed{t \approx 1.633}$$

Step 6: Degrees of freedom

For a one-sample T-test:

$$df = n - 1$$

Therefore:

$$df = 16 - 1$$
$$\boxed{df = 15}$$

Step 7: Critical value

For:

$$\alpha = 0.05$$

and:

$$df = 15$$

for a two-tailed test. Refer the table of t-test

$$t_{critical} \approx \pm 2.131$$

Our test statistic is:

$$t = 1.633$$

Since:

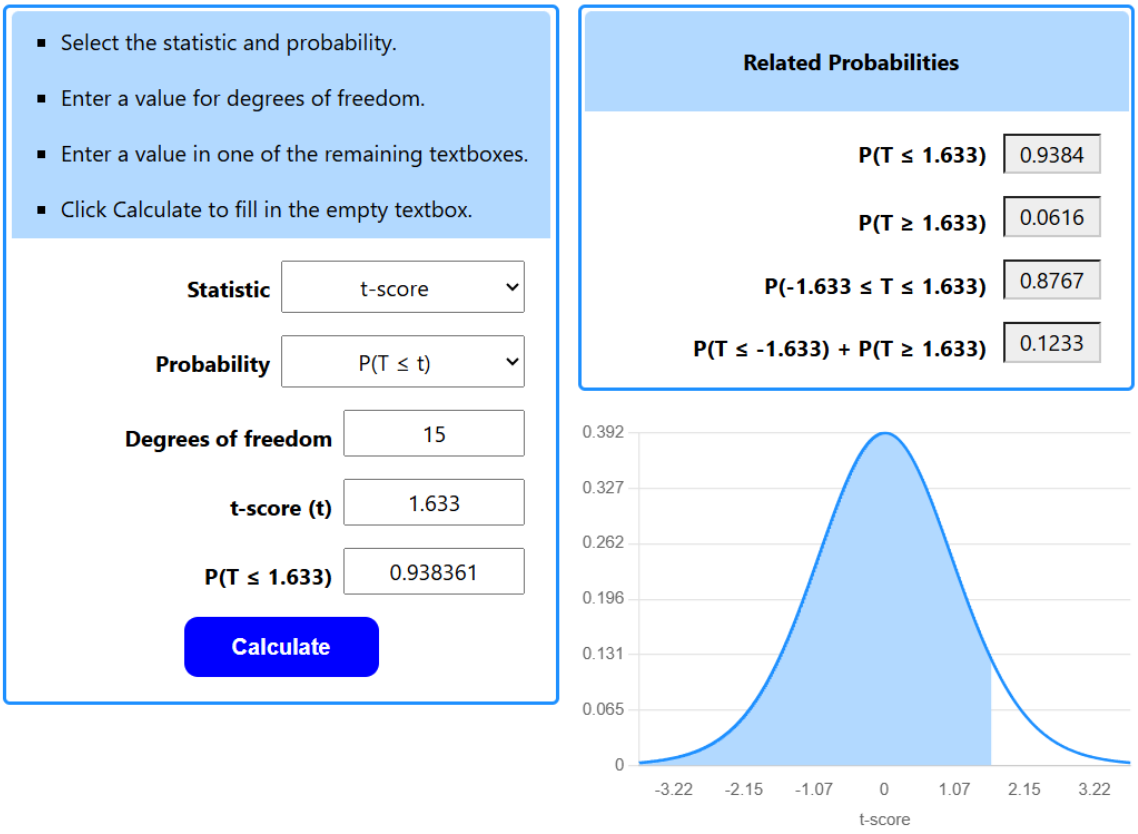
$$| 1.633 | < 2.131$$

we **fail to reject** H_0.

Step 8: Calculate p-value

The two-tailed p-value is approximately:

$$p \approx 0.123$$



Compare:

$$0.123 > 0.05$$

Therefore:

$$\boxed{\text{Fail to reject } H_0}$$

Conclusion

At the 5% significance level, there is **not sufficient statistical evidence** to conclude that the average customer waiting time is different from 20 minutes.